

Metaphysical necessity, numerical identity, and the a priori

PHIL UA-78 Metaphysics

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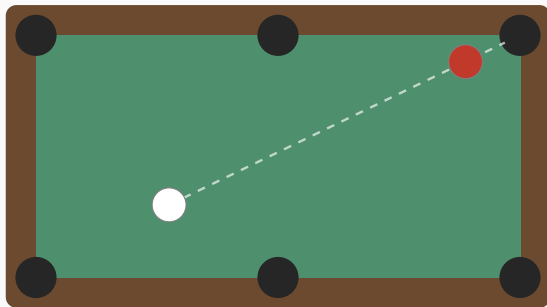
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Metaphysical necessity and possibility

Different interpretations of 'could have' in English

'Could have' is flexible: on different occasions, we can use it to mean different things.

- ▶ The white ball couldn't have gone into the pocket without hitting the red ball (since the red ball was right in front of the pocket)
- ▶ Someone could have picked up the red ball and made the white ball go straight into the pocket without hitting any other balls.



Broader and narrower interpretations

Some of these meanings are **broader** than others.

- The claim that interpretation 1 of 'could have' is broader than interpretation 2 means, roughly, that when a sentence 'It could have been that ...' is true on interpretation 2, it is also true on interpretation 1 (holding fixed the interpretation of whatever goes in the dots).

'Could have' and 'had to'

'Could have' goes hand in hand with 'had to'.

- ▶ 'It could have not been that P ' $\approx$ 'It didn't have to be that P '
- ▶ 'It had to not be that P ' $\approx$ 'It couldn't have been that P '

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And hence...

- ▶ 'It could have been that P ' $\approx$ 'It didn't have to not be that P '
- ▶ 'It had to be that P ' $\approx$ 'It couldn't have not been that P '

We say that one interpretation of 'had to' is broader than another when the interpretation of 'could have' corresponding to the former is broader than the interpretation corresponding to the latter.

- The claim that interpretation 1 of 'had to' is broader than interpretation 2 means, roughly, that when a sentence 'It had to be that ...' is true on interpretation 1, it is also true on interpretation 2 (holding fixed the interpretation of whatever goes in the dots).

Necessity and possibility

'Could have' is one of many ways we have in English for talking about (a wide range of notions of) *possibility*, and 'had to' is one of many ways of talking about (the corresponding range of notions of) *necessity*.

- ▶ 'It could have been that P ' $\approx$ 'It could be that P ' $\approx$ 'It is/was possible for it to be that P ' $\approx$ 'Possibly, P '...
- ▶ 'It had to be that P ' $\approx$ 'It must be that P ' $\approx$ 'It is/was necessary that P ' $\approx$ 'Necessarily, P '...

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The general pattern:

- ▶ Possibly $\approx$ not necessarily not
- ▶ Necessarily $\approx$ not possibly not

This is called the logical *duality* of 'possibly' and 'necessarily'.

Nomic and metaphysical modality

In this course we will be especially interested in *two* of the many available interpretations of modal words like 'possible' and 'necessary'.

- 1 **Nomic** (aka **physical**) necessity and possibility. Here, 'It could have been that P ' can be restated less ambiguously as 'The laws of nature are compatible with its being that P ', and 'It had to be that P ' can be restated as 'The laws of nature guarantee that P '.

2 Metaphysical (aka **absolute**) ne-

cessity and possibility. This is what Kripke calls 'necessity and possibility *tout court*'. *[I will argue] that characteristic theoretical identifications like 'Heat is the motion of molecules', are not contingent truths but necessary truths, and here of course I don't mean just physically necessary, but necessary in the highest degree, whatever that means. (Physical necessity, might turn out to be necessity in the highest degree. But that's a question which I don't wish to prejudge. At least for this sort of example, it might be that when something's physically necessary, it always is necessary tout court.) (N&N p. 99).*

Many philosophers—including me—are happy to characterize metaphysical modality as the *broadest* modality. That is:

- ▶ If there is *any* sense in which things ‘could have been’ a certain way, then it’s metaphysically possible for them to be that way.
- ▶ If it’s metaphysically necessary that things are a certain way, then it’s true in every sense that they ‘had to be’ that way.

Nomic and metaphysical modality

You may be more familiar with the expressions ‘logically possible’ and ‘logically necessary’, which some philosophers used to use for what’s now called ‘metaphysical’ possibility and necessity. But this can be misleading.

- It is (plausibly) metaphysically necessary that no bachelor is married. But the **sentence** ‘No bachelor is married’ is not provable in any known system of logic—in logic, any two nouns or adjectives are interchangeable.

One would standardly say that ‘No bachelor is married’ isn’t a *logical truth*. But this remark just concerns the *sentence*, not what it says about the world.

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Philosophers sometimes confuse things by saying ‘It is not logically necessary that no bachelor is married’ when they mean “‘No bachelor is married’ is not a logical truth’. But this is bad practice—in philosophical writing, one needs to be a stickler about using quotation marks.

One reason to care about this topic

Recall that last week we were talking about “comprehensive world-pictures” like Democriteanism, which include a hard-to-make-sense-of claim ‘completeness’ (which I wrote as ‘that’s all’).

The Democritean’s idea is that if you say the truth about what atoms there are, what shapes they are, and how far apart they are (at every time), you have in some sense given a ‘complete’ description of the world. But what could this *mean* (if it’s not obviously false)?

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A plausible idea is that at least *part* of what it means is a **supervenience** claim. *Every fact* is metaphysically necessitated by the facts about atom-arrangement: that is, every fact is such that it’s metaphysically necessary that if the atom-arrangement facts are as they are, that fact obtains.

Numerical identity and Leibniz's Law

The concept of numerical identity

‘This is numerically identical to that’ $\approx$ ‘This and that are one and the same’ $\approx$ ‘This is the very same thing as that’ $\approx$ ‘This is that’.

There’s another use of ‘identical’ in play when we talk about two twins, or two peas in a pod, or two copies of a newspaper, or two electrons being ‘identical’. For two things to be ‘identical’ in this other sense—‘qualitatively identical’—is for them to be *exactly similar*, at least in some relevant range of respects.

By contrast, it simply can’t happen that two things are numerically identical! If x is numerically identical to y , then x and y are not two things, but one thing.

Sometimes, a claim about numerical identity can be equivalently expressed using numerals:

- ▶ 'I own a cat x and a cat y such that x is not numerically identical to y ' $\approx$ 'I own at least two cats'
- ▶ 'For any cats x and y , if x is owned by me and y is owned by me, then x is numerically identical to y ' $\approx$ 'I own at most one cat'.

Some terminological notes

- 1: In this course, when we just say 'identity'/'identical' without modification, we always will mean numerical, never qualitative.
- 2: 'Distinct' means 'not (numerically) identical'
3. At least in philosophy (and arguably also in mathematics), the 'equals' sign '=' is almost always used to mean 'is numerically identical to'. ' $\neq$ ' means 'is not numerically identical to'.

4. In this course we will always be using the English word ‘identity’ interchangeably with ‘being identical’—compare ‘adjacency’/‘being adjacent’. ‘Identity’ also has a different use as a count noun, as in ‘Eating eight apples a day is an important part of Mike’s identity’. This use will play no role in the course. If you want to use the word ‘identity’ in your writing in a way that couldn’t be replaced with ‘being identical’, probably something has gone wrong.

The logic of numerical identity

There are two basic logical principles about numerical identity.

Reflexivity Everything is identical to itself.

Leibniz's Law If x is identical to y , y has every property that x has.

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We can deduce two other useful principles from these

Symmetry If x is identical to y , y is identical to x .

Transitivity If y is identical to x and z is identical to y , z is identical to x .

The logic of numerical identity

Proof of Symmetry: Suppose x is identical to y . By Reflexivity, x is identical to x ; or in other words, x has the property of being identical to x . But by Leibniz's Law, y has every property that x has. So, y also has the property of being identical to x , or in other words, y is identical to x .

The logic of numerical identity

Proof of Symmetry: Suppose x is identical to y . By Reflexivity, x is identical to x ; or in other words, x has the property of being identical to x . But by Leibniz's Law, y has every property that x has. So, y also has the property of being identical to x , or in other words, y is identical to x . *Proof of Transitivity:* Suppose y is identical to x and z is identical to y . By Reflexivity, x is identical to x ; or in other words, x has the property of being identical to x . By Leibniz's Law, y has every property that x has and z has every property that y has, which implies that z has every property that x has. So, z has the property of being identical to x ; or in other words, z is identical to x .

An obviously atrocious argument against Transitivity

(a) and (b) are both true, but (c) is false:

(a) Kamala Harris is numerically identical to some human being

(b) Some human being is numerically identical to Donald Trump

(c) Kamala Harris is numerically identical to Donald Trump

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Obviously this is not a good objection to Transitivity. We can't infer from (a) and (b) that there is something such that Kamala Harris is identical to it and it is identical to Donald Trump.

An obviously atrocious argument against Transitivity

There's a general moral here. In English, **determiner phrases** like 'some human being' and 'a human being' look syntactically like proper names ('Kamala Harris') and pronouns ('she', 'it'). But *logically*, they behave in rather differently. For example, for many such determiner phrases, 'For any object x , x does such and such' is compatible with '[DP] does not do such-and-such'.

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This also applies—although less obviously—to determiner phrases using 'the', of the form 'the F ' (e.g. 'the president of the US').

Bogus counterexamples to Leibniz's Law

(Pointing to a photo showing a new car on the dealer lot.) This car is shiny. *(Pointing to a photo showing a beat up old car.)* That car is not shiny. So, this car is not numerically identical to that car.

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(Pointing to a photo showing a new car on the dealer lot.) This car is shiny. *(Pointing to a photo showing a beat up old car.)* That car is not shiny. So, this car is not numerically identical to that car. Obviously something has gone wrong here! I could just as well have said 'This car is shiny and was once owned by me' and 'That car is not shiny and was once owned by me'. Given 'This car is not numerically identical to that car', I could infer 'There are x and y such that x is not numerically identical to y and x was once owned by me and y was once owned by me', and hence 'there are at least two cars that were once owned by me'.

Bogus counterexamples to Leibniz's Law

Diagnosis: when I say 'This car is shiny' and 'that car is not shiny' I mean something like 'This car was shiny at the time this photo was taken' and 'That car was not shiny at the time that photo was taken'. Or maybe I mean 'This car is depicted as shiny in this picture' and 'That car is depicted as not shiny in that picture'. Anyhow, *my two uses of 'is shiny' express two different properties.*

So, I cannot infer from 'This is shiny and that is not shiny' to 'There is a property that this has and that does not have', for exactly the same reason that I can't infer from 'That animal (pointing at one animal) is a cat and that animal (pointing at a different animal) is a dog' to 'There is an animal that is a cat and is also a dog'.